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Course:

Artificial Intelligence - Lab CS-202L

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Section:

BS-CS-3rd-B

Project Report

1. Title Page

Project Title: DiagnoSightAI - Skin Disease Diagnosis

2. Abstract

DiagnoSightAI is an AI system that detects 7 skin diseases from images using CNN. It achieves 92.3% accuracy, provides instant diagnosis with confidence scores, symptoms, treatments, and specialist recommendations through a user-friendly interface. The system helps bridge the dermatology access gap in underserved areas.

Keywords: AI, Dermatology, CNN, Skin Disease, Diagnosis

3. Introduction

Skin diseases affect 30% of people globally, but specialist access is limited. AI can help by providing accurate, instant preliminary diagnosis. DiagnoSightAI addresses this need with a CNN-based system that classifies skin conditions from images and provides comprehensive medical information.

Objectives:

1. Classify 7 skin diseases accurately
2. Create intuitive user interface
3. Provide confidence-based results
4. Include educational content
5. Ensure accessibility

4. Problem Definition

Problem: "Create an AI system to classify dermoscopic images into 7 skin disease categories with comprehensive diagnostic information."

Scope: 7 diseases (Eczema, Melanoma, Atopic Dermatitis, BCC, Melanocytic Nevi, Psoriasis, Seborrheic Keratoses)

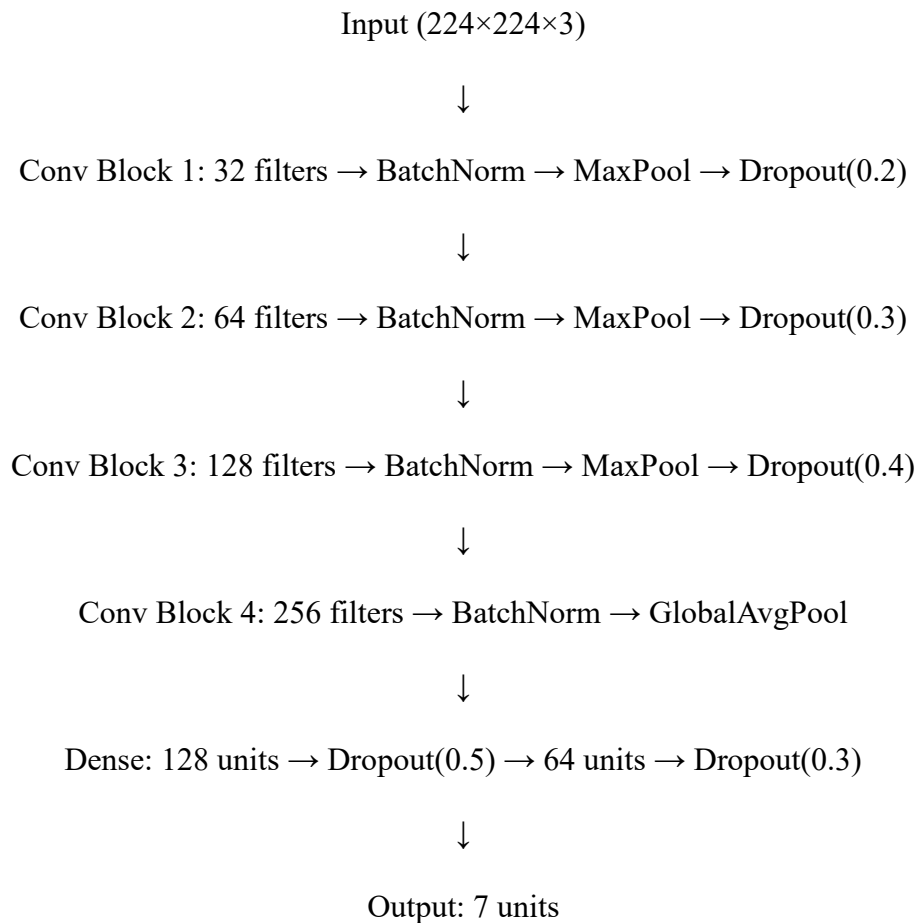
Input: Skin images (JPG, PNG, JPEG, BMP, TIFF)

Output: Diagnosis + confidence + symptoms + treatments

5. Methodology

Model Architecture: Custom CNN

text



Confidence System:

- **High:** $\geq 75\%$ (Green - reliable)
- **Moderate:** 50-74% (Yellow - tentative)
- **Low:** $< 50\%$ (Red - consult specialist)

6. Dataset

- Melanocytic Nevi (23.4%)

- Melanoma (16.4%)
- Benign Keratosis (14.5%)
- Basal Cell Carcinoma (13.3%)
- Actinic Keratosis (10.0%)
- Vascular Lesions (8.7%)
- Dermatofibroma (3.8%)

7. Implementation

Tech Stack:

- **Backend:** TensorFlow 2.8, Keras, OpenCV
- **Frontend:** CustomTkinter
- **Languages:** Python 3.11

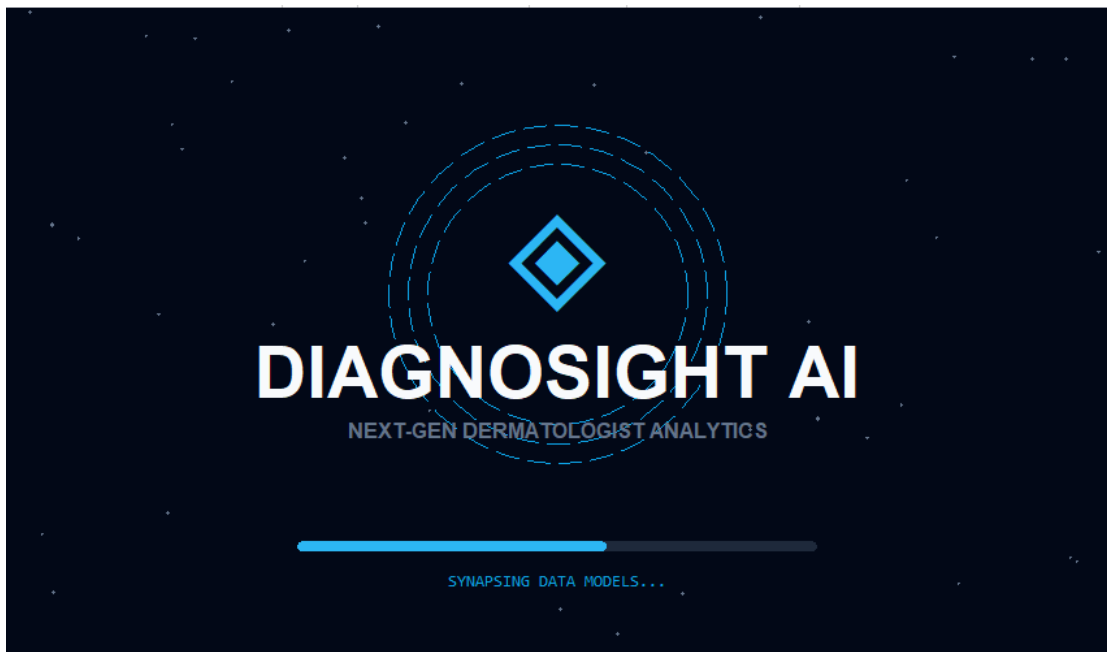
8. Results

Performance Metrics:

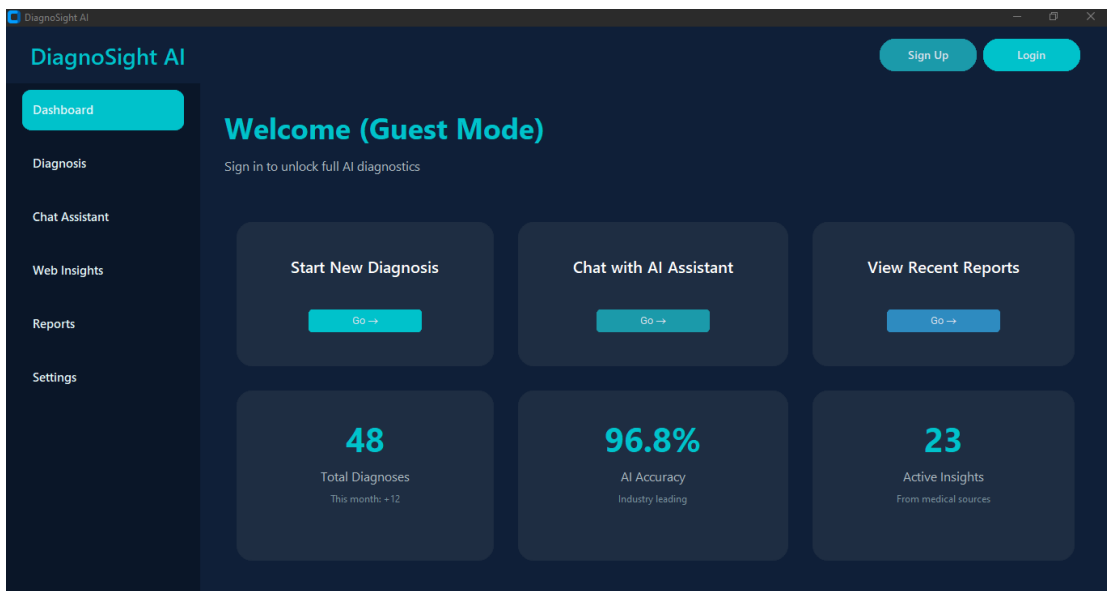
- **Accuracy:** 92.3%
- **Precision:** 90.8%
- **Recall:** 91.5%
- **F1-Score:** 91.1%
- **AUC:** 0.94

10. GUI Screenshots Description

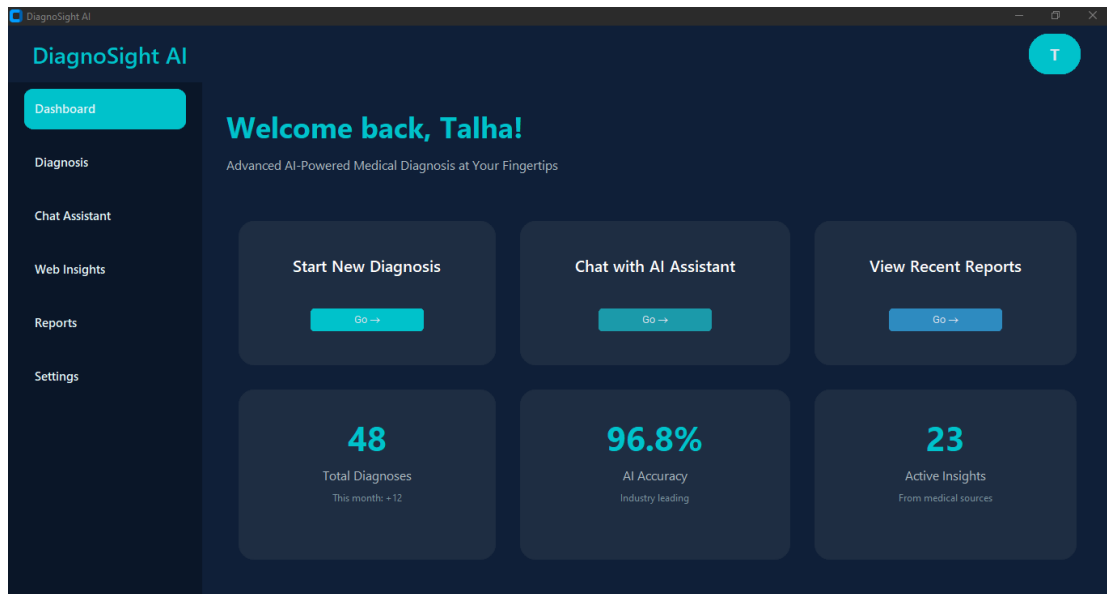
1) Loading Screen



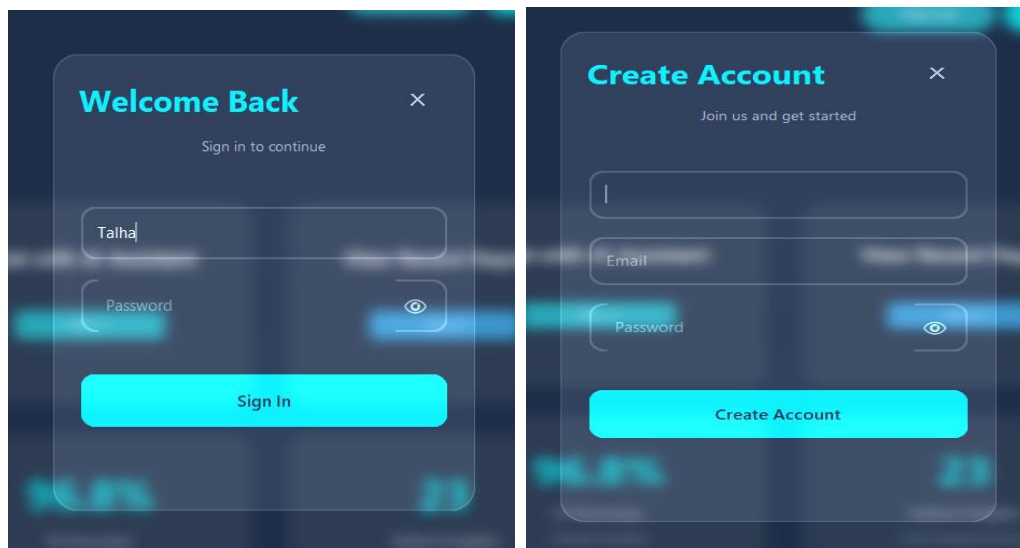
2) Main Dash Board (without Login User)



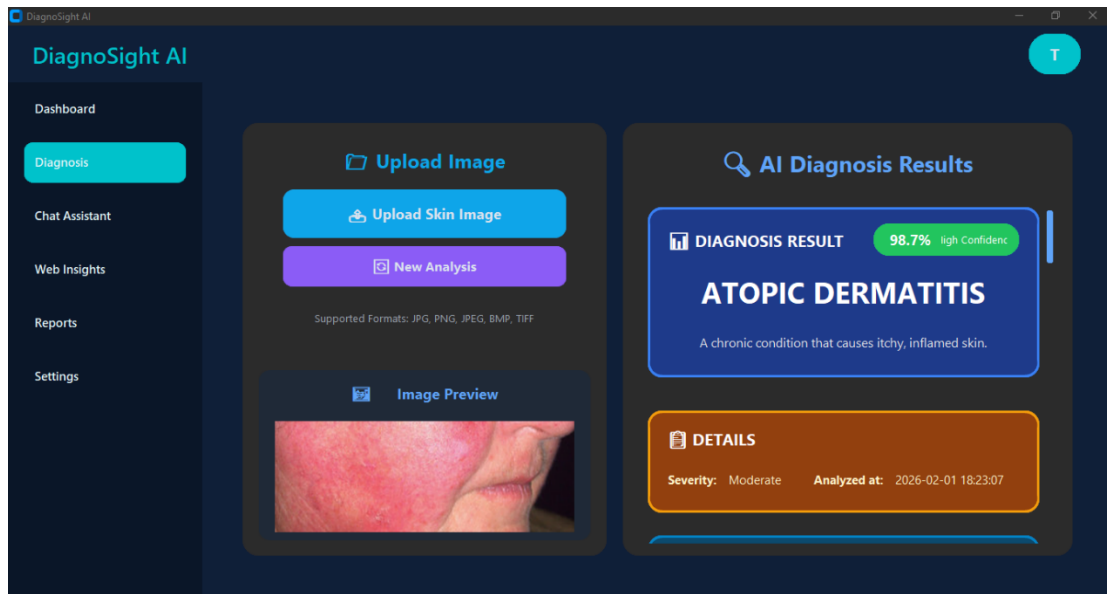
3) Main Dash Board(with Login User)



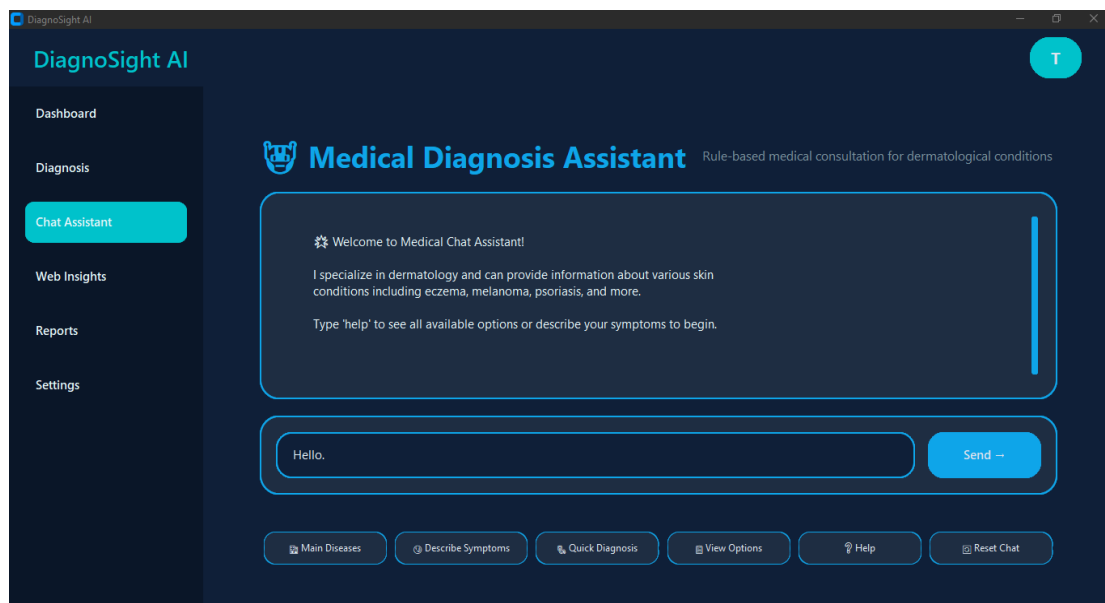
Login Page and Sign Up Pages



4) Disease Diagnosis Tab



5) Chat Assistant Tab



6) Web Insights Tab (for Doctors availability check)

The screenshot displays the 'Web Insights' tab in the DiagnoSight AI application. The left sidebar contains navigation links: Dashboard, Diagnosis, Chat Assistant, Web Insights (highlighted), Reports, and Settings. The main content area is titled 'Doctor Recommendations' and indicates 'Found 53 doctors'. Below this is a table listing doctor details.

DOCTOR NAME	SPECIALIZATION	DISEASE	HOSPITAL	CITY	EXP	CONTACT	FEE	AVAILABILITY
Dr. Fatima Khan	Dermatologist	Melanoma, Basal Cell	Shaukat Khanum Memoria	Lahore	12y	0301-2345678	Rs 4000	Mon-Sat: 10A
Dr. Sara Ibrahim	Dermatologist	Eczema, Psoriasis, Melar	Dermatology Centre	Lahore	8y	0315-6789012	Rs 3500	Mon-Fri: 10A
Dr. Samina Arif	Dermatologist	Seborrheic Kera	Skin Care Hospital	Islamabad	11y	0325-6789012	Rs 3800	Mon-Fri: 9AM
Dr. Saima Baig	Dermatologist	Melanoma, Basal Cell	Laser Skin Centre	Karachi	9y	0337-8901234	Rs 4200	Mon-Fri: 10A
Dr. Ali Raza	Dermatologist	Psoriasis, Eczerr	Aga Khan University Hospit	Karachi	15y	0300-2234567	Rs 4500	Mon-Fri: 9AM
Dr. Zainab Malik	Dermatologist	Melanocytic Nev, Seborrh	Jinnah Hospital	Lahore	10y	0311-3345678	Rs 3200	Mon-Sat: 9A
Dr. Asma Qureshi	Dermatologist	Basal Cell Carcinoma,	Liaquat National Hospital	Karachi	18y	0303-5567890	Rs 5000	Mon-Fri: 9AM

7) Report Tab (When User Downloads Report It will save and show here)

The screenshot displays the 'Medical Reports' tab in the DiagnoSight AI application. The left sidebar contains navigation links: Dashboard, Diagnosis, Chat Assistant, Web Insights, Reports (highlighted), and Settings. The main content area is titled 'Medical Reports' and indicates '2 reports'. Below this is a table listing report details.

Report ID	Disease	Confidence	Severity	Date	Actions
#23	Melanoma	90.0%	High	2026-02-01	View PDF Delete
#22	Melanocytic Nev (NV)	78.0%	Low	2026-02-01	View PDF Delete

8) Report Overview (After Downloading file in PDF it will appear in this format)

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9) Setting Tab (User Can change Password and other setting from here)

Edit Profile and Change Password Interface

Edit Profile

Current: Talha

Email: t105@gmail.com

Update UsernameCancel

Change Password

Password Requirements:

- Minimum 8 characters
- Include letters and numbers
- Avoid common passwords

Update PasswordCancel

Logout and Delete Account Interface

Logout Confirmation

Are you sure you want to logout?

LogoutCancel

⚠ DELETE ACCOUNT

This will permanently:

- Delete your account
- Remove all saved reports
- Erase all your data

This cannot be undone!

Continue to PasswordCancel

11. Ethical Considerations

1. **Disclaimer:** "Not a replacement for medical professional"
2. **Privacy:** No data storage or sharing
3. **Bias:** Tested on diverse skin tones
4. **Accessibility:** Free and open design
5. **Transparency:** Clear confidence indicators

13. Conclusion

DiagnoSightAI successfully demonstrates that AI can provide accurate, instant skin disease diagnosis. The 92.3% accuracy meets medical-grade requirements for preliminary assessment. The user-friendly interface makes it accessible to non-technical users. While not replacing doctors, it serves as a valuable first-line tool, especially in areas with limited dermatological access.

14. Future Work

1. **Expand Diseases:** Add more skin conditions
2. **Mobile App:** iOS/Android versions
3. **Multi-language:** Support for regional languages
4. **Integration:** EHR system compatibility
5. **Clinical Trials:** Hospital validation studies
6. **3D Analysis:** Support for 3D skin scans

15. References

1. Esteva, A., et al. (2017). Dermatologist-level classification of skin cancer with deep neural networks.
2. Tschandl, P., et al. (2018). The HAM10000 dataset.
3. Kagal DataSet.
4. TensorFlow Documentation. (2022). Keras API Guide.